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University-industry-government linkages and the helix theory on the fourth industrial revolution

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Abstract

In the current era of the fourth industrial revolution, the creation of new knowledge and the production and diffusion of innovation constitute the most critical dimensions of development and under-development. In this direction, for over two decades, a useful conceptual contribution is the triple-helix theory: the interconnection of universities, firms, and governmental policies. The aim of this study is through a periodization of the helix theory literature to understand how this approach is related to new perceptions of innovation and to describe its possible future analytical perspectives. Within the present phase of the fourth industrial revolution, the institutional dimensions of a socioeconomic system (including universities, industries, and government policies) are following complex and co-evolving development trajectories and we must perceive them in their specific historical and spatial configurations.

Keywords: helix theory, socioeconomic development, the fourth industrial revolution, university-industry-government linkages, local development

1. Introduction: Positioning the issue

In the current phase of crisis and restructuring of globalization, the scientific discourse that concerns all socioeconomic disciplines seems to be in a phase of substantial repositioning (Ghorra-Gobin, 2012; Laudicina & Peterson, 2016; Rodrik, 2011; Sapir, 2011). The focus seems to shift from the perspective of traditional economic geography and regional development to the study of local dynamics, as the latter approach can synthesize the individual spatial levels into a global perspective (Asheim, Grillitsch, & Trippl, 2016; Vlados, Deniozos, Chatzinikolaou, & Digkas, 2019). On a deeper level, a relatively unseen “paradigmatic change”—in Kuhnian terms (Kuhn, 1962)—is gradually taking place, where the methodologies and interpretations of all the interdisciplinary approaches, from the whole spectrum of socioeconomic sciences, are being repositioned and resynthesized.

Today, this paradigmatic change appears in the “Fourth Industrial Revolution” thematic, which has also been adopted quickly by formal international institutions such as the World Economic Forum (Schwab, 2016). In a neo-Schumpeterian perception (Hanusch & Pyka, 2007; Levinthal, 2006; Perez, 2010; Witt, 2008), an industrial revolution can be any radical socioeconomic change that takes place and gets structurally assimilated over time (Köhler, 2012). Specifically, the Fourth Industrial Revolution refers to the current, rapidly advancing, technological progress of humanity, as the latest developments mainly in the fields of robotics and artificial intelligence can increase the overall productivity of society drastically (Park, 2018; Peters, 2017). Some analysts relate this historical phase to the emergence of the Cyber-physical systems (Colombo, Karnouskos, Kaynak, Shi, & Yin, 2017; Davis, 2016; Schlick, 2012): these systems result from a new and more integrated production process that is based on decentralized network systems and processes (Bloem et al., 2014). Often, there is also reference to the effects of the Fourth Industrial Revolution on future employment (Dombrowski & Wagner, 2014; Manu, 2016; World Economic Forum, 2016), where creativity is of crucial importance for all socioeconomic actors (Gray, 2016).

According to Maynard (Maynard, 2015), this new phase requires substantial investments in partnerships between the public and the private sector. Otherwise, the gap between the poor and the rich will deepen, the geopolitical conflicts will intensify, and the sustainable development initiatives will be undermined; if we do not succeed as a whole to respond to these challenges, then our world is going to face a recurrence of the events that marked the emergence of previous industrial revolutions.

In the related analyses, however, the fact that the overall socioeconomic development is a matter of “cellular” competitiveness is usually neglected; and by cellular we mean the micro-competitiveness which is generated by the organizational syntheses at strategic, technological and managerial level (Vlados, Deniozos, Chatzinikolaou, & Demertzis, 2018a; Vlados, Katimertzopoulos, & Blatsos, 2019). In other words, the organic competitiveness, which, if conceived in a unifying approach, it is structured synthetically at all the analytical levels (micro-meso-macro), refers to the ability of any socioeconomic organization/system to innovate and manage the produced change effectively (Dopfer, 2011; Peneder, 2017).

In today’s world of crisis and restructuring of globalization, knowledge production and transfer are crucial for all socioeconomic organizations. In the seminal work of Gibbons et al. (Gibbons, 2000; Gibbons et al., 1994), a new Mode of knowledge production has emerged (Mode 2), which is interdisciplinary and different from the traditional Mode 1 of knowledge production. The Mode 1 has reached its interpretative limits, including social values, concepts, and methods, rules and standards which were established by the Newtonian (or positivist) model and imposed its criteria of validity and scientific practice. On a second level, this relates to the need of perceiving the concept of crisis differently nowadays, by moving towards a structural and evolutionary perspective instead of “conjunctural” (Vlados, Deniozos, Chatzinikolaou, & Demertzis, 2018b).

In this context, the evolutionary linkages of universities-industries-governments, as approached by the helix theory, are crucial for the production of new knowledge. In the past mode of knowledge production, the only institutions that were responsible for generating knowledge were the universities with their internal research mechanisms. However, it seems nowadays that in order to innovate effectively—and ultimately survive—as a socioeconomic organization, it is critical to produce and assimilate new knowledge. By applying an evolutionary/biological perspective (Geus, 2002; Kennedy, Miller, & Niewiarowski, 2018; McMullen, 2018; Meyer & Davis, 2003), where all spatialized socioeconomic systems are co-determined and co-evolve, it is clear that the triple helix institutions (university-industry-government) are not separate functional spheres, but inseparable and strongly related institutional entities. Therefore, in the contemporary scientific discourse of innovation, the co-evolving triple system of helices starts to gain significant popularity.

Under the prism of these methodological concerns, the present study attempts to explore the evolution of helix theory in the literature, from its theoretical foundations to this day. The aim is to understand how this approach is related to a new perception of innovation within the present phase of the fourth industrial revolution. To achieve this goal, this article presents a literature review of contributions to helix theory, separated into three successive phases: (a) phase of theoretical foundation (1995-2000), (b) phase of conceptual expansion (2001-2010), and (c) phase of recent developments and systematic implementation attempts (2011-2018). Finally, this study attempts to find some useful analytical future developments and enrichments to the helix theory.

2. The evolution of helix theory

2.1. The phase of theoretical foundation (1995-2000)

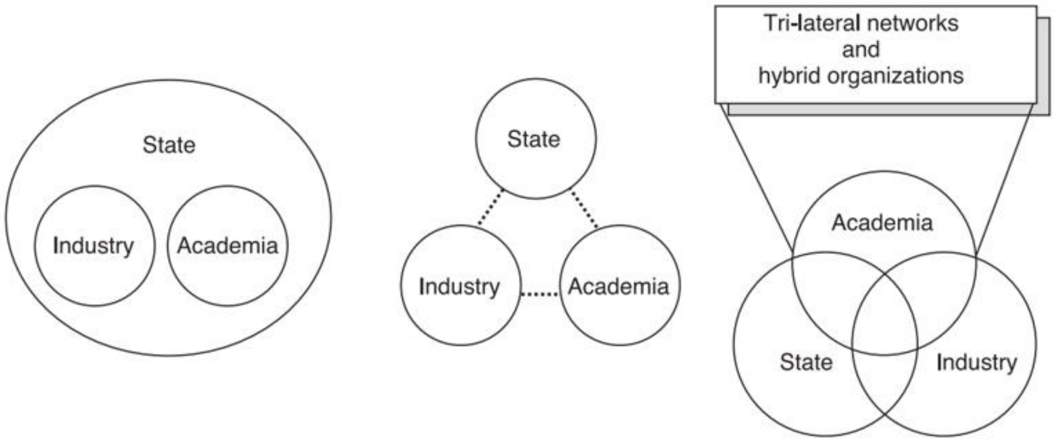
i. Etzkowitz and Leydesdorff (1995) introduce the Triple Helix model as a “laboratory for knowledge-based economic development.” According to the authors, the universities and industries are now involved in tasks that were formerly mostly the province of the other. In this context, the governments are offering incentives and pressing academic institutions to move beyond the traditional functions of the cultural memory, education, and research, and have a more direct contribution to the creation of wealth.
ii. Etzkowitz (1996) argues that a hidden assumption of evolutionary economics is the development of the theory of the firm: this focus is sometimes broadened to a group of companies in an industry or a heterogeneous network of companies in an industrial district. He proposes to extend this analysis, especially, at the regional level, in order to incorporate governmental and academic actors, who have an essential role in creating the conditions for a thriving regional innovation environment.
iii. Specifically, in this point of view, the development of academic research capacities will define future economic and social development in the form of human capital, tacit knowledge, and intellectual property. Consequently, to transfer knowledge flows into new sources of technological innovation is now an academic task that transforms the traditional structure and function of the university. This potential unfolds through organizational innovations such as technology transfer offices, incubator facilities, and research centers with industrial participation. (Etzkowitz & Leydesdorff, 1998a)
iv. Etzkowitz and Leydesdorff (1998b) relate the Triple Helix System to the concept of “endless transition.” According to the authors, the term “economies in transition" was primarily used for the Middle and Eastern Europe nations, because they suffered from a crisis caused by the global transition to a knowledge-based economy. Nowadays, it the Eastern European economies are not the only ones in this phase of transition, because the complex dynamics of social relations between institutionalized spheres are increasingly “locked into” a regime of technological innovation and organizational reform for a multiplicity of socioeconomic systems.
v. According to Leydesdorff and Etzkowitz (1998), the Triple Helix conceptual framework is a recursive model of how an overlay of communications operates on the underlying institutions. This overlay corresponds to the market selections, the innovative dynamics, and the network controls which provide different codes of communication at the global level. “Local translations” at the interfaces induce adaptation mechanisms in the institutional arrangements. The authors argue that when two dynamics tend to coevolve into trajectories, a regime of transitions emerges when trajectories can be recombined and, therefore, the emerging hyper-networks are expected to be in flux. In this context, the institutions can be flexible in temporarily assuming the roles of other partners, while crucial is now the role of niche management and human capital management.
vi. Viale and Ghiglione (1998) suggest that we can interpret the Triple Helix model, one the one hand, as a “neo-corporatist” approach that focuses on reaching a consensus among the actors of academia, industry, and government, with the involvement of “innovation coordinators.” On the other hand, the evolutionary Triple Helix model focuses on specific local contexts of universities, government, and industry that are learning to encourage economic growth through “generative

relationships" that may resemble coupled reciprocal relations that persist over time. These reciprocal relations cause changes in the way agents conceptualize their environment and how to act in it.

vii. Leydesdorff (2000) describes the triple helix as an evolutionary model of innovation by using simulations to demonstrate how lock-in can be enhanced using coevolution like the one between regions and technologies. However, he argues that the third source of random variation into the system, such as the triple helix system, may reverse in a later stage the system and lead to more complex arrangements of market segmentation. Therefore, a mechanism for "lock-out" can also be introduced in this theoretical approach.

viii. The phase of foundation ends with the contribution of Etzkowitz and Leydesdorff (2000) who argue that we can compare the Triple Helix of "university-industry-state" with alternative models for explaining the current research system in its social contexts, such as "Mode 2" and National Systems[1]. They argue that new technologies cause increasingly new reorganizations across industrial sectors and nations states. Therefore the consequent transformations can be interpreted in terms of neo-evolutionary mechanisms and procedures. They conclude that a triple-helix arrangement that reorganizes the knowledge infrastructure in terms of possible overlays can be expected to be generated endogenously. In this context, the research taking place in academia can act as a locus in the ''laboratory'' of such knowledge-intensive network transitions. Furthermore, they illustrate how a triple helix system differs from the "etatistic" (state-oriented) or the "laissez-faire" (market-oriented) model because it includes tri-lateral networks and hybrid organizations (see Figure 1).

Figure 1: From "etatistic" and "laizzez-faire" to triple helix. Adapted from Etzkowitz and Leydesdorff (2000)



2.2. The phase of conceptual expansion (2001-2010)

ix. Baber (2001) argues that the specific nature and dynamic of the triple helix of state-industry-university is related to the impact of globalization and the overall institutional reconstruction and

[1] A National System of Innovation refers to the flow of technology and information between different socioeconomic actors (people, enterprises and institutions), which is crucial for the development of innovation, at the national level. This theory, which originates in the works of mainly Freeman (Freeman, 1987) and Lundvall (Lundvall, 1992) in the late 1980s, suggests that technology and innovation development is the outcome of complex relationships between the actors in the national system, specifically between enterprises, universities and government research institutes.

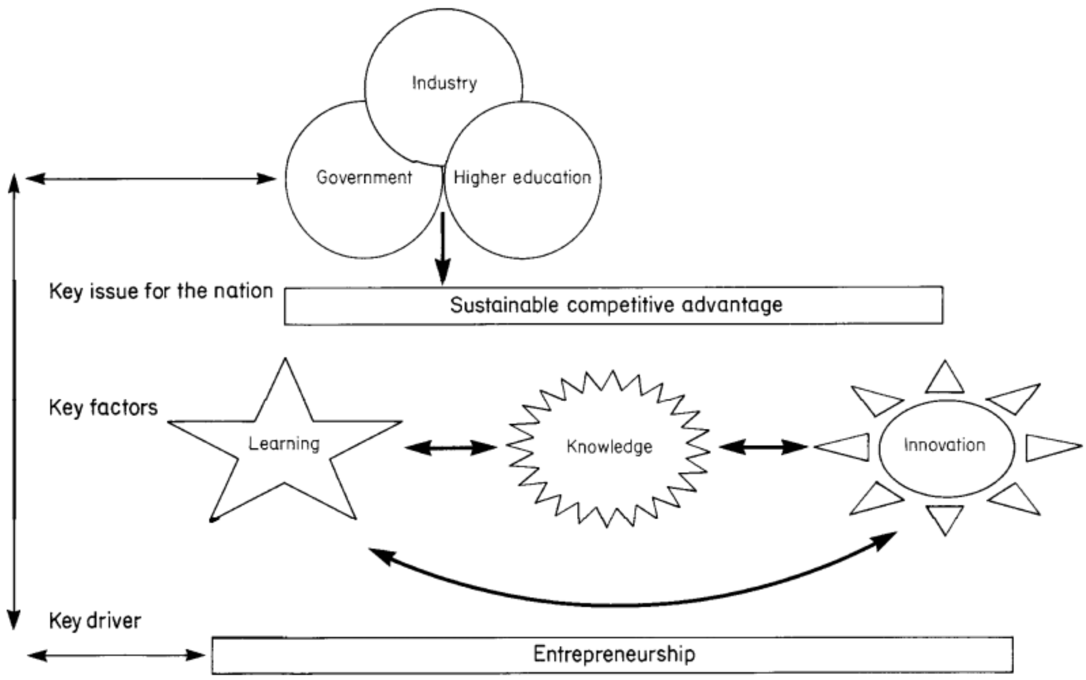
the emergence of trans-disciplinary scientific fields. Overall, the author discusses the impact of these transformations for the debates over the emerging knowledge society. The author concludes that the predicted demise of capitalism due to the emergence of knowledge societies is an ideological hyperbole, but it is crucial to understand the fundamental forces that are dramatically restructuring institutions inherited from an earlier era.

x. Shinn (2002), with a critical perspective, after analyzing specific citations and Internet search data to explore the “geographical locations, audiences and impact,” comments that the body of work pioneered over the first few years by the triple helix theory and other approaches to the new production of scientific knowledge has specific strengths and weaknesses. The author suggests that the challenge is to ensure that these new approaches do not nourish an unproductive “frenzy” but, on the contrary, to contribute something enduring to scholarship and practice.

xi. Viale and Campodall'Orto (2002) extend the distinction of the Triple Helix into the two different versions, that is the evolutionary and neo-corporatist, by arguing that in the USA there is a strong ability for using the scientific knowledge in the industry as compared to the EU. The US system of R&D is focused on industry, private finance, and the market, while the respective EU system focuses on public financing, agencies, and public programs for technological innovation and transfer. Therefore, they conclude that where the evolution of a strong interaction of “university-industry” is shaped by regulation and market forces—and not by the direct intervention of government and public bodies—is then able to produce the best science and technology output.

xii. Matlay and Mitra (2002) examine the case of the triple helix with particular reference to the UK, arguing that the concept of helix theory is useful to understand the central role of sustaining competitiveness in entrepreneurship and learning. According to the authors, entrepreneurship is the key driver among the key players’ relationships within a national system (see Figure 2).

Figure 2: Relationships between the key players. Reproduced from Matlay and Mitra (2002)



xiii. Leydesdorff and Etzkowitz (2003) examine whether “the public” can be considered as the fourth helix, arguing that if we position the public as the fourth helix, then this is narrowed into another private sphere. They argue that the civil society is the foundation of the enterprise of innovation and the ability of individuals and groups to express themselves freely without permission from the state is a necessary condition for the development of a triple helix dynamics. In this context, the authors refer to the example of Brazil, when the military lost its power, and large-scale technology programs failed, and then university science and technology researchers were able to encourage the systematic creation of start-ups and incubators.

xiv. Etzkowitz (2003) notes that the increased importance of knowledge and the role of academia in the incubation of technology-based firms has given it a more prominent place in the institutional firmament. Therefore, the author suggests that the "entrepreneurial university" should embrace its proactive role in the dissemination and creation of new knowledge while, in this context, it functions according to an interactive rather than a linear model of innovation. Accordingly, the firms nowadays are raising their technological level, and they gradually approach the academic model, while the governments act as public entrepreneurs and venture capitalists in addition to their traditional regulatory role. The triple helix model ignites a model of "innovation in innovation" and, in advance, that innovation, beyond product development, has now become an endogenous process of "taking the role of the other," encouraging the hybridization among the institutional spheres.

xv. The innovating region, according to Etzkowitz and Klofsten (2005), is a model of knowledge-based regional development conceptualized as a set of multi-linear dynamics and different technological paradigms. By using longitudinal data from a Swedish region, and international comparisons, they distinguish four stages of innovative regional development: inception, implementation, consolidation, and renewal. They conclude that innovation policy is materialized bottom-up as an outcome of collective entrepreneurship through business, government, and university collaborations.

xvi. Viale and Etzkowitz (2005) analyze the historical changes to the relationship of tacit to codified knowledge over three industrial revolutions. They argue that knowledge is now increasingly "polyvalent" because new theoretical, practical, and interdisciplinary implications form a common center of gravity. In this context, the entrepreneurial university is the epicenter in the ongoing "Third Academic Revolution."

xvii. According to Etzkowitz and Zhou (2006), instead of adding a fourth helix, we can conceptualize Triple Helix as a set of two triple helices or Triple Helix twins. They suggest that the introduction of a fourth helix, such as civil society, might lead the triadic model of helix theory to lose its dynamics. Therefore, they argue that an expanded model is required and propose a "Sustainability Triple Helix" of university–public–government as a complement to the Innovation Triple Helix of "university-industry-government"; in this context a missing element is introduced into the system, while the triple helix model is able to retain the dynamic properties of a "tertius gaudens".

xviii. Huggins et al. (2008) stress the importance of knowledge for regional innovation and development, particularly noticing the theoretical emergence of higher education institutions as a base for the commercialization of knowledge. By exploring a lagging region in Wales, they conclude that it is difficult for the universities to become the bases of "commercializable" knowledge and, therefore, Wales might have to establish publicly-funded research institutes complementary to the activities of universities.

xix. Saad et al. (2008) point out that the policy shift in developing countries, which gives gradually to universities the role of being the sustainable basis for innovation and technological progress, is nowadays critical. They use the examples of Malaysia and Algeria to demonstrate that the triple helix model should act as a guideline to enhance the role of universities in developing countries as agents of innovation and sustainable development.

xx. Johnson (2008) analyzes the role of Precarn, an intermediating organization in Canada—an independent non-profit organization focusing on the pre-commercial development of leading-edge technologies—and argues that this case study can facilitate the understanding of how triple-helix collaborations unfold.

xxi. Zhou (2008) presents the case study of a successful entrepreneurial university in China (Northeastern University), concluding that government-pulled and industry-university collaborations can initiate the pathway to an entrepreneurial university. The author supports this argument with the successful software company (Neusoft) created in China's first science park.

xxii. Bjerregaard (2009) examines the micro-dynamics in the evolution of strategic cooperation between universities and small and medium-sized enterprises, by using a qualitative study of the total population of university departments and SMEs in Denmark, involved in collaborative research projects sponsored by a new governmental program. There are different strategic orientations, whether short or long-term, that differentiate the procedures and the results of academia-industry collaborations accordingly.

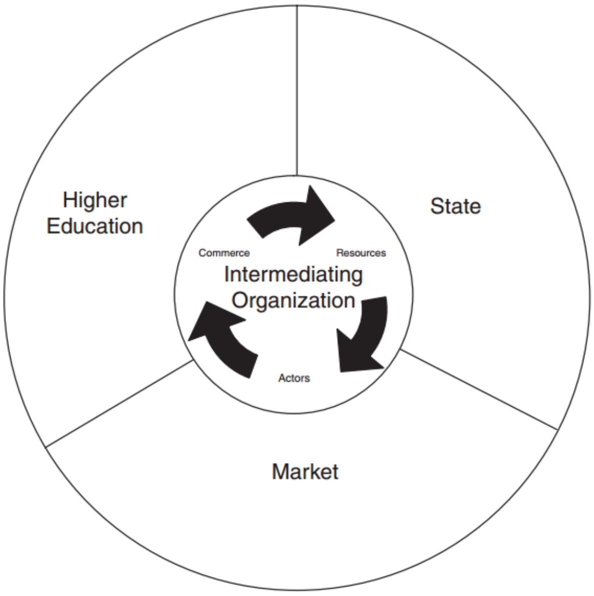
xxiii. Etzkowitz and Ranga (2010) argue that spaces (instead of spheres) of knowledge, innovation, and consensus, whose forms can provide guidelines for integrating both endogenous and exogenous strategies, can define a knowledge-based regional development system. The concept of spaces is an attempt to understand the dynamics of the triple helix in terms of policy articulation based on knowledge-based regional innovation systems. The knowledge-based regions—in particular, like Silicon Valley—took decades to develop, under conscious interventions and institutional formation and reformation.

xxiv. In the developing economy of Thailand, Yuwawutto et al. (2010) study how an intermediary organization (Industrial Technology Assistance Program (ITAP)) can strengthen the SME’s innovation, by linking university knowledge to the industry. They conclude that technology policies based on the Triple Helix model, when complemented by intermediary agencies such as ITAP, can foster the competitiveness of SMEs.

xxv. According to Viale and Pozzali (2010) in order to explore in more depth the theoretical and empirical basis of the triple helix, the literature focusing on “complex adaptive systems” can offer useful suggestions. On this basis, every actor in a social system plays a role within a grid of relationships that are either active or that could be activated; therefore, any change in the way an actor’s incentives are structured will generate actions that may affect his or her social network.

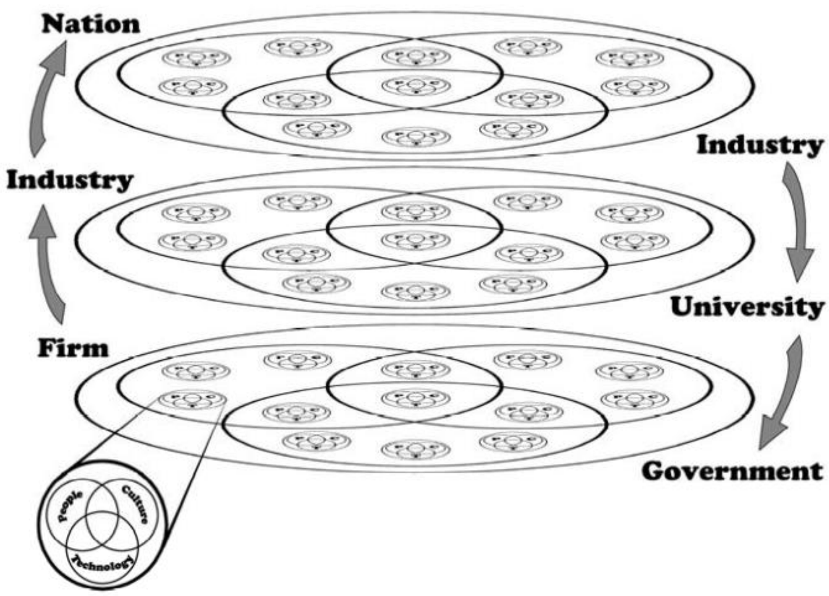
xxvi. Metcalfe (2010) examines the trilateral networks of the triple helix by using the case of two North American intermediary organizations. The author argues that the intermediary organizations, which are often nonprofit organizations, influence the formation of the triple helix relations through the exchange of actors, resources, and commerce. The intermediating organizations are the points (nodes) between public and private entities (see Figure 3).

Figure 3: Model of an intermediating organization. Reproduced from Metcalfe (2010)



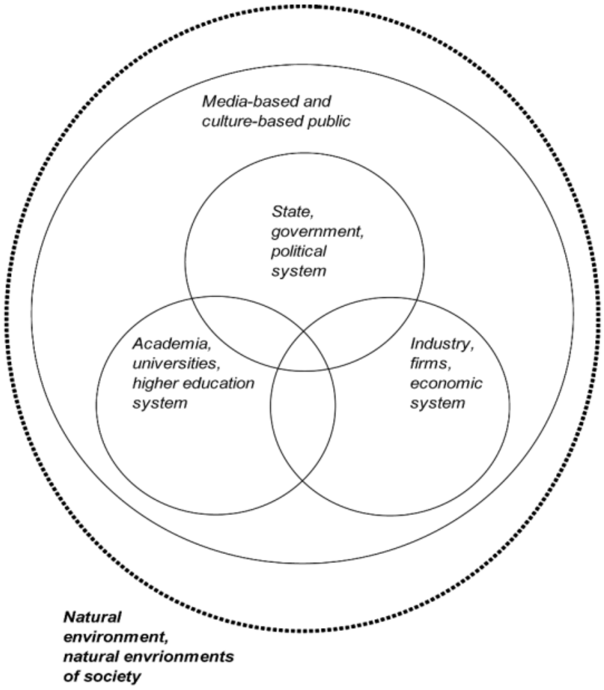
xxvii. Carayannis and Campbell (2009) argue that a new mode regarding the production of knowledge is underway: that is, the Mode 3 knowledge production of the co-existence and co-evolution of different knowledge and innovation paradigms. In this context, they introduce the Quadruple Helix that integrates the perspective of the media-based and culture-based public, which is an emerging fractal innovation ecosystem well-configured for the knowledge economy and society. The authors argue in particular that there are “knowledge fractals,” which emphasize the continuum-like bottom-up and top-down progress of complexity (see Figure 4).

Figure 4: The 21st-century fractal innovation ecosystem. Reproduced from Carayannis and Campbell (2009)



xxviii. Furthermore, Carayannis and Campbell (2010) introduce the Quintuple Helix. They suggest the existence of an inter-disciplinary and trans-disciplinary framework of analysis of knowledge, innovation, and the environment (natural environments). Therefore, they introduce a five-helix structure model, namely the Quintuple Helix. They describe this model as related to sustainable development and social ecology where the concepts of “eco-innovation” and “eco-entrepreneurship” are better suited for today’s knowledge society (see Figure 5).

Figure 5: The five-helix model of the Quintuple Helix. Reproduced from Carayannis and Campbell (2010)



2.3. The phase of recent developments and systematic implementation attempts (2011-2018)

xxix. According to Leydesdorff and Deakin (2011), the evolutionary model of Triple Helix allows us to investigate the level of knowledge of an urban economy about the support for innovation provided by the social factor of a city (the civil society). This cultural development is not a spontaneous procedure deriving from market interactions, but the outcome of governmental interventions, academic initiatives and quality, and business strategies, all of which have to be carefully managed and articulated for the requirements of an urban regeneration program.

xxx. Etzkowitz (2011) emphasizes, in this step, the role of the university in creating new ways for socioeconomic development in the transition to a knowledge-based society. This new role for the university—sometimes called the “third mission”—requires fundamental changes at all the organizational and institutional levels of the triple helix in order to create an innovation environment based on science, technology, and entrepreneurial spirit.

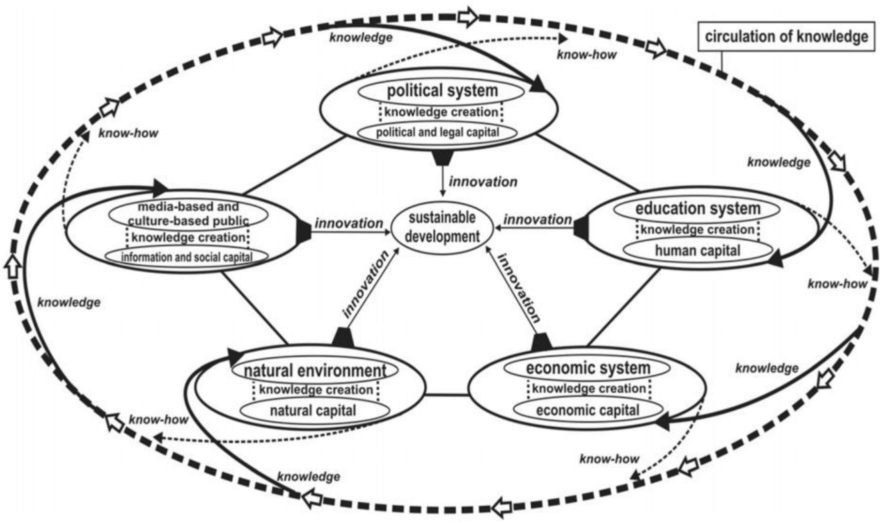
xxxi. Marcovich and Shinn (2011) propose four modifications to the triple helix innovation model. First, the introduction of a fourth strand, namely society. Second, to refer to the strands as binomials, that is, doublets of university/society, university/industry, industry/society, and other related doublets. Third, to conceive the binomials in a hierarchical form (for example, in the university/society strand university may be dominant and the society secondary). Fourth, helix theory processes can be temporary segmented phases. Finally, they support this argument by using the case study of “Dip-

Pen nanolithography", and by identifying four phases, characterized by specific binomial accompanied by a hierarchy: (a) academic instrument research (university/society), then the transition phase from instrument to tool follows, (b) company start-up (university/industry), (c) the mature firm and commercialization (industry/society) and (d) confirming the societal strand ''nanofication'' (society/industry).

xxxii. Carayannis and Campbell (2011) highlight the concept of "democracy of knowledge" that refers to the knowledge and innovation heterogeneity and diversity in the advanced economy and society. They introduce the concept of open innovation diplomacy that can bridge the distance and other divides (cultural, socioeconomic, technological) with focused and properly targeted initiatives in order to connect ideas and solutions with markets and investors ready to operationalize them. In this direction, "open innovation diplomacy" can be a novel strategy, policy-making, and governance approach in the context of the quadruple and quintuple innovation helices.

xxxiii. Furthermore, Carayannis et al. (2012) claim that the Quintuple Helix refers to the necessary socioecological transition of society and economy in the twenty-first century and, according to the authors, "the Quintuple Helix is ecologically sensitive." The Quintuple Helix supports here the formation of a win-win situation between ecology, knowledge, and innovation. That seems able by creating synergies between the economy, society, and democracy. In this context, the issue of global warming represents an area of ecological concern to which the Quintuple Helix innovation model has a potential application, by placing at the center the concept of sustainable development (see Figure 6).

Figure 6: Quintuple-helix model and its functions. Reproduced from Carayannis et al. (2012)



xxxiv. Nakwa et al. (2012) explore the roles of innovation intermediaries in stimulating triple helix networks in the SMEs of Thailand's developing economy. They collect data from the SME-based ceramic and furniture industries to test the significance of the ways to intermediate the triple helix process. They find that the role of intermediaries is vital for promoting the development of triple helix network in the case of these industries.

xxxv. Consequently, Fogelberg and Thorpenberg (2012), by studying the policies for regional innovation in Sweden, they mainly analyze the development of innovation policy and development organizations known as "Arenas." According to the authors, these organizations were modeled and function according to the triple helix innovation theory. After analyzing the historical development of two such "Arenas" created by public and private actors in two Swedish cities, conclude that diverging interests—between the expectation of an ideal-type of Triple Helix process and the actors' view of the actual development process—may result in unresolved tensions within Triple Helix Arenas, diminishing their impact drastically.

xxxvi. Leydesdorff (2012) stresses that one may wish to move beyond three helix environments, but also a fourth or fifth dimension would require substantive specification and operationalization in terms of potentially relevant data, and sometimes the further development of relevant indicators. The author concludes that as long as there is no operationalization in the relatively simple case of three dimensions, one should be cautious in generalizing beyond the triple helix model to an N-tuple of helices. In addition, he explains why he stopped co-authoring with H. Etzkowitz commenting that the tension between the two models—Etzkowitz' as neo-institutional one focusing on interinstitutional networking and exchanges and his neo-evolutionary theoretical framework examining mechanisms of exchanges among functions (wealth creation, knowledge production, and normative control)—provided heuristics for a number of years. Although, after the co-authored articles in the early 2000s, they decided to stop coauthoring because of possible misunderstandings and confusions about the two models and their graphical representations.

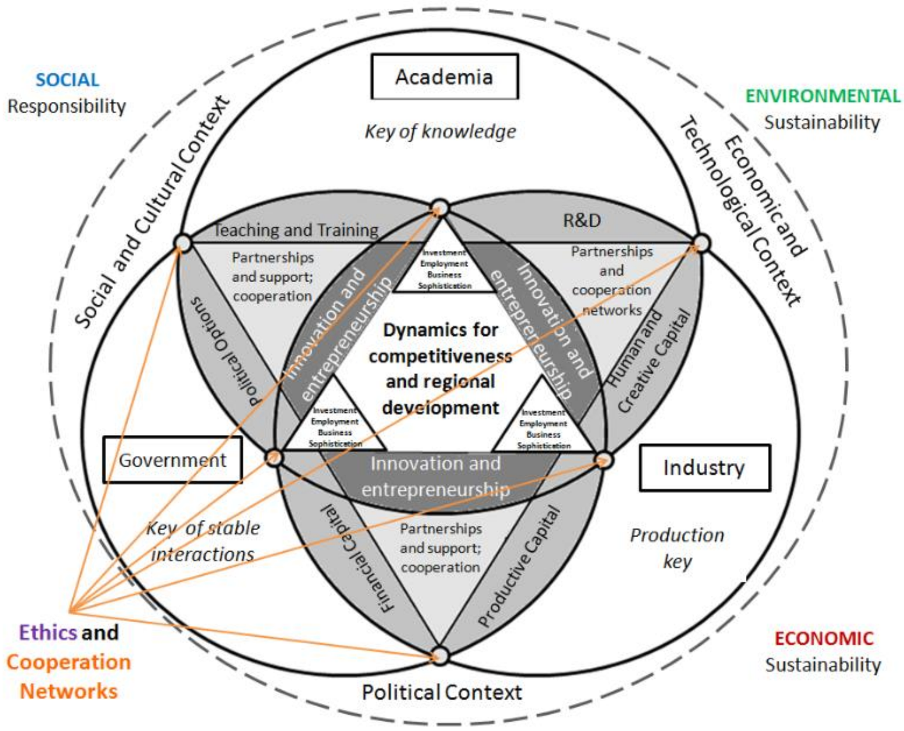
xxxvii. Yang et al. (2012) explore what the triple helix frameworks can offer to the analysis of eco-innovation dynamics, by comparing four triple helix frameworks in terms of outset, characteristic and analytical focus (see Figure 7). The triple helix twins framework can be suitable for stressing tensions in industry and public relationships, while the quadruple helix and N-tuple helices frameworks are suitable for analyzing the co-production of innovation beyond the "university-industry-government" relationships, as stressed in the original triple helix model. Also, concerning eco-innovation dynamics, the authors argue that the concern for common environmental benefits calls for adding NGOs and local communities as a fourth actor in the helix of innovation: the quadruple and N-tuple helix frameworks are suitable for this purpose.

Figure 7: Outset, characteristics, and analytical focus for different helix models of innovation. Reproduced from Yang et al. (2012)

	Outset	Characteristic	Analytical focus
Triple helix	Enhancing the role of universities in innovation	Places a particular emphasis on tri-lateral networks and hybrid organizations: university–industry–government	Knowledge infrastructure unstable tri-lateral networks
Triple helix twins	Introduce a missing element (public) into the model, while retaining the dynamic properties of a *tertius gaudens*	Emphasizes a need for public participation to ensure innovation is not harming environment and health: innovation triple helix of university–industry–government; Sustainability triple helix of university–government–public	Tensions between industry (presumed economic interest) and public (presumed social and environmental interest)
Quadruple helix	Public reality is crucial for a society to assign top-priorities to innovation and knowledge	Emphasizes that culture and media-based public is as important as university, industry and government to be the fourth actor in innovation	Co-evolution and cross-integration of different knowledge modes
N-tuple helices	Complexity of innovation and diversity of actors	Break state, market and civil society into as many subsystems as possible	A plurality of agents, actors and organizations are involved

xxxviii. Farinha and Ferreira (2013) propose a new conceptual model which displays a dynamic triple helix framework able to clarify the importance of innovation and entrepreneurship as factors of competitiveness and regional development. The Triple Helix Triangulation model features a new model that leverages the triple helix dynamics by focusing on local innovation and entrepreneurship as catalyzers of development and a region's ability to compete globally (Figure 8).

Figure 8: Triple Helix Triangulation model. Reproduced from Farinha and Ferreira (2013)



xxxix. Rodrigues and Melo (2013) note that the triple helix model is used as a source of policies and programs to foster innovation. They question whether the triple helix model provides a conceptual basis for development policies, particularly at the local level. They, therefore, explore the case of a small Portuguese municipality where the development policy effort is not only guided by the model but targets at the materialization of local triple helices. They find that owing to deficiencies in each of the three helices and the linkages between them, some territories and localities do not have the required resources to emulate the trajectories of localities such as Silicon Valley.

xl. Ranga and Etzkowitz (2013) introduce the concept of “triple helix systems” as an analytical construct that synthesizes the critical features of triple helix model interactions into an “innovation system” formation. They define the “triple helix systems” according to systems theory as a set of components, relationships, and functions. They conclude that from a triple helix systems perspective, the articulation and the non-linear interactions between the spaces can generate new combinations of knowledge and resources. These interactions can advance innovation theory and practice, especially at the regional level.

xli. Van Rijnsoever et al. (2014), by using insights from organizational learning, attempt to formulate hypotheses about the “micro-incentives” for knowledge development in the Triple Helix. They explore how the combinations of declarative and procedural knowledge bases influence the publication and patent output by using data about publications, patents, and funding in the domain of carbon capture technology. They find that the incentives to engage in triple helix knowledge-based development derive from the organization's own prior declarative and procedural knowledge bases, but not much from collaboration with other parties.

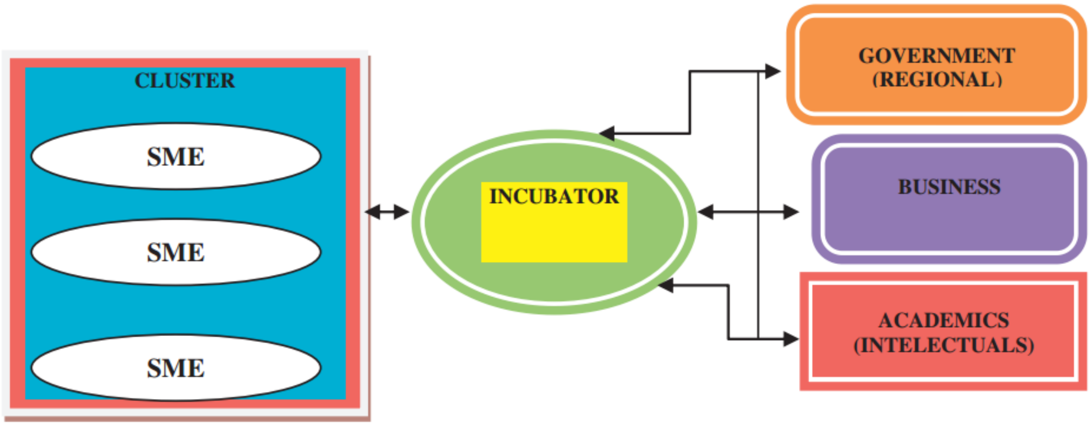
xlii. According to Cai (2014), the triple helix concept has been developed from the experience of advanced economies in the West, and therefore the author stresses a lack of theoretical considerations and empirical evidence on whether the triple helix model is applicable in non-

Western contexts and less developed socioeconomic systems. By examining the “institutional logic” of China compared to Western societies, the author suggests that in order to optimize the Chinese innovation policies, China needs to adapt its institutional environment to facilitate the interactions between key innovation actors and to develop its triple helix modes according to the unique Chinese institutional environment.

xliii. Smith and Leydesdorff (2014) argue that we require new conceptual tools and methodologies in order to understand how economies change by interactions with science and government. By applying the evolution of the metaphor of a Triple Helix into an evolutionary model, they explore how triple-helix relationships are both continuing and mutating and the conditions under which a triple helix model might be seen to be unraveling in the face of pressures on each of the three helices. They find that the “footlooseness of high technology manufacturing and knowledge-intensive services counteract the embeddedness prevailing in medium technology manufacturing.” Therefore, conclude that the geographical level at which synergy in Triple Helix relations can be expected and sustained varies among nations and regions.

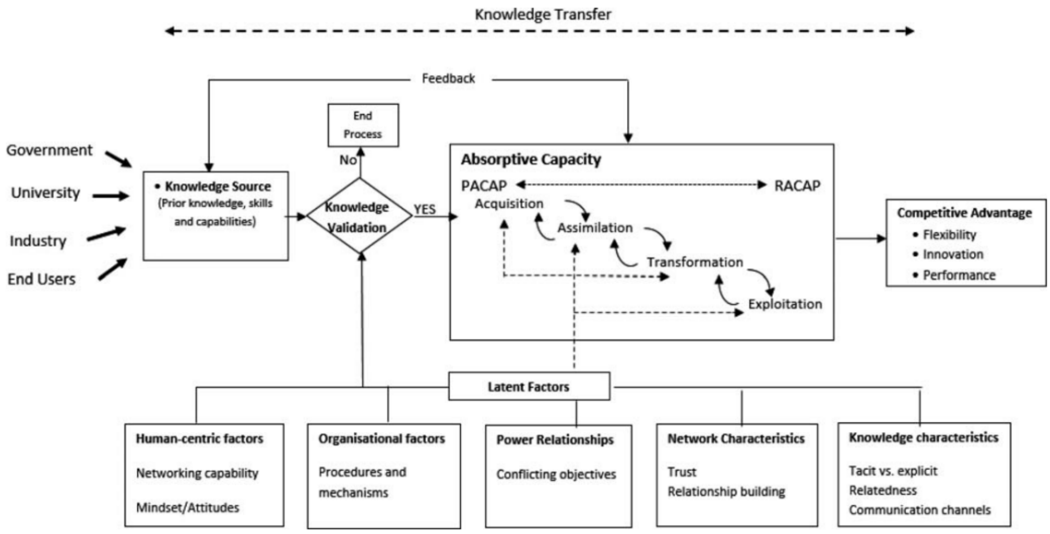
xliv. According to Herlian (2015), the SMEs contribute significantly to the national economy, and a “regional innovation cluster” approach is useful to enhance their competitiveness. This innovation derives from the successful collaboration between academia, the industry, and the regional government. The author argues that regional cluster development and the growth of SMEs have to be supported by innovation-based business incubation activities (see Figure 9).

Figure 9: Regional innovation clusters of SME: Triple Helix concept. Reproduced from Herliana (2015)



xlv. Miller et al. (2016) examine how knowledge transfer from universities to the broader regional knowledge ecosystem offers opportunities for increased regional innovation and commercialization. They try to improve, precisely, the understanding of knowledge transfer in an open innovation context where multiple diverse quadruple helix stakeholders are interacting, by proposing an absorptive capacity-based conceptual framework. After document analysis and longitudinal observation data over a three-year period, and by using an exploratory methodology with semi-structured interviews, they identify five factors which mediate both the ability of stakeholders to participate in knowledge transfer and the effectiveness of knowledge acquisition, assimilation, transformation and exploitation: human-centric factors, organizational factors, knowledge characteristics, power relationships and network characteristics (see Figure 10).

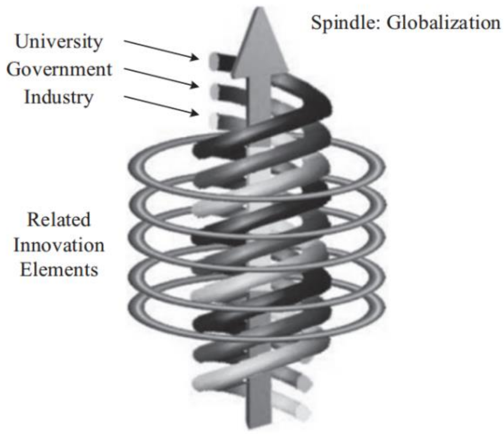
Figure 10: Ex post absorptive capacity based conceptual framework for knowledge transfer from universities. PACAP stands for potential absorptive capacity while RACAP for realized absorptive capacity. Reproduced from Miller et al. (2016)



xlvi. Kimatu (2016) focuses on the strategic aspect of triple-helix interactions and argues that the innovative and sustainable economic development of a country relies upon the mutual interaction for strategic objectives between the triple-helix model institutions. Within the developed nations, that has given rise to science parks, “technopolis,” and at more advanced stages to “innopolis” (that is, the innovations in these countries have significantly increased because of these mutual interactions and have made significant technological cities called technopolis and later innopolis). By supporting the necessity of a healthy civil society in the triple helix, concludes that it is imperative for the developing and middle-income countries to learn and implement the discussed global best practices of science park creation in the triple helix settings.

xlvii. Cheng et al. (2017) argue that nanotechnology is a critical technology that also has an essential influence on the emergence and development of a series of high-technologies in the 21st century and that is because of its multidisciplinary nature. Nanotechnology, in this point of view, requires extensive multi-sector collaborations to foster more effective development outcomes. According to the authors, a triple helix model to analyze the globalization of China’s nanotechnology innovation can include three aspects. First, the independent and interactive relationship among universities, governments, and industries. Second, the vertical progress of globalization. Third, some influential, innovative elements throughout the process. They demonstrate the model by studying the case of the China International Nanotech Innovation Cluster. They assume that the propelling roles and powerful impact three main entities played in the nanotechnology globalization process, as well as the trend and challenge of globalization, offer practical implications for the triple helix innovation in other industries. With respect to globalization and triple helix, they stress that globalization has brought the speedy movement of capital, information, technology, talent, and other critical, innovative elements and, therefore, as the leading entities in the process of globalization all governments, universities, and industries should take advantage of the influential, innovative elements flowing in from abroad (see Figure 11).

Figure 11: A novel triple-helix model. Reproduced from Cheng et al. (2017)



xlviii. Grundel & Dahlström (2016) attempt to understand and describe the possible preconditions for the transformation of a regional innovation system into a quadruple and quintuple helix system applied to the development of a sustainable forestry-based “bioeconomy” in Värmland, Sweden. They find that the use of a quintuple helix RIS in this locality for the transformation to a forestry-based bioeconomy could be a possible way forward towards sustainability. Concretely, they propose that the involvement of civil society in the innovation system could contribute to a more significant societal transformation that aims to change consumer behavior, production patterns, technological developments, infrastructure, norms and values, and the overall socioeconomic environment.

xlix. Wonglimpiyarat’s (2016) research reviews the role of the innovation incubator and the “university business incubator” in supporting the entrepreneurial development in Thailand. The author defines the university business incubator as an incubator established by the university to provide office space, equipment, mentoring services as well as other administrative support to assist the formation of new ventures. The author examines the case studies of leading university business incubators and science and technology incubators and suggests that an incubation program is one of the critical policy mechanisms to support innovation. Proposes that university business incubators should act as an intermediary between the spheres of university and industry to provide interactive linkages and contribute to the effective utilization of university research.

l. Sarpong et al. (2017) review the discursive practices in order to explore how the organizing practices of the industry, university and government facilitate (or impede) developing countries transition to a hybrid triple helix model of innovation. They identify three domains of practices: a) advanced research capabilities and external partnerships, b) the quantification of scientific knowledge and outputs, and c) “collective entrepreneurship,” that constitutively facilitate (or impede) partnership and in turn the successful transition to a hybrid triple helix model. They also suggest that this conceptual framework also highlights the contextual influence of differential schemata of interpretations on how to organize innovation by the three institutional actors in developing countries.

li. Altaf et al. (2018) examine the role of the “Offices of Research Innovation and Commercialization” in Pakistan in order to find whether these offices facilitate triple Helix interactions. They apply a qualitative research method with semi-structured interviews, and they conclude that Pakistan does not have a triple helix system, claiming that their research results can assist “ORIC established universities” and linked firms to overcome the key issues and challenges faced by them in collaboration. They suggest that universities and firms can overcome the challenges through better coordination and communication and that the monitoring role of government and the role of other organizations are crucial.

lii. Sá et al. (2018) explore the triple helix dynamics in the context of rural entrepreneurship and the entrepreneurs’ perceived contributions to local development, by studying the case of a peripheral region in Portugal. They find that although the local government had an essential role by fostering and supporting an entrepreneurial culture, the knowledge-based regional development is not a set of top-down policies, but rather a complex local discovery process involving multiple actors. In particular, they stress that although previous studies assume that the Triple Helix fosters technological innovation that favors regional development, mainly by adopting a macro-level perspective, this study contributes to the micro-level dynamics of the triple helix model, through the view of low tech, rural entrepreneurs, considering their context.

liii. Ryan et al. (2018), in a micro-level perspective, analyze how the triple helix model is able to give an “organizing regime” within which participant firms can extend their capabilities for explorative innovation through funded collaborations with academia, by studying a qualitative case study of a set of research projects that comprise the “Centres for Science, Engineering and Technology” project in Ireland. In particular, the authors decomposed the “micro-foundations” of the processes, interactions, and the structures that facilitated the extension of member firms’ explorative innovation capabilities nurtured within each university-industry partnership. They find that by using a micro-foundational lens, they can have a deeper understanding of how triple helix programs influence the capabilities of firms for explorative innovation.

liv. From a similar theoretical perspective, Cunningham et al. (2018) argue that the conceptual models of the quadruple helix have primarily taken a macro perspective and they fail to perceive the value creation activities at the micro level of the quadruple helix. By focusing on the way value is collectively created, captured, and enhanced at the micro level of the quadruple helix, they develop a micro level conceptual model of “principal investigators” as value creators in the quadruple helix.

lv. Höglund and Linton (2018) explore the case of Robotdalen, a smart specialization initiative in the region of Mälardalen, Sweden, and its impact on regional innovation systems. This regional initiative was set up to create a regional center for robotics based on the triple helix innovation systems model of industry, university, and government interaction. To understand the quadruple helix models’ complexity and, after utilizing a longitudinal perspective of 10 years (2007–2016) and exploring the micro activities from a strategizing perspective, they identify strategic practices that have evolved for this locality.

3. Conclusions and implications: The helix theory as a new perspective of innovation in the fourth industrial revolution

After this synoptic literature review and from the different historical phases of the helix theory (theoretical foundation/conceptual, expansion/recent developments, and systematic implementation attempts) it seems that we are still on the early operational stages of the theory. Nevertheless, the helices system is useful to understand better how innovation is created and disseminated.

In this context, the concept of endless transition to a knowledge-based economy is indeed one of the most useful contributions of the triple-helix theory (Etzkowitz & Leydesdorff, 1998b). With a more radical point of view, Loet Leydesdorff (2018), one of the helix theory founders, suggests that the “fourth industrial revolution” concept introduces a new metaphor which can perhaps replace older models, such as “national systems of innovation”, the “knowledge-based economy”, and the “triple helix of university-industry-government relations”. The author argues that the fourth industrial revolution, as a pending transition, it is driven by technological innovations.

It seems, therefore, that the helix theory can frame particularly useful interpretations for the upcoming events of the fourth industrial revolution, where “global hyperconnectivity” (Kelly, 2019) and computer applications are going to flood even further our daily routines. However, technological innovation is not something exogenous to the socioeconomic system, as some technologically-focused approaches imply. Innovation can only emerge as an evolutionary synthesis of strategy, technology, and management that the various organizations have the potential to build and valorize (Vlados, Deniozos, Chatzinikolaou, & Demertzis, 2018a; Vlados, Katimertzopoulos, & Blatsos, 2019). The fourth industrial revolution, which is a historical and evolutionary phase of global capitalism and which brings various subversive dimensions for any kind of socioeconomic organization (businesses or universities or governments), will probably unfold as a new accumulation regime and open the next phase of the global economy (Aglietta, 2010; Boyer, 1986).

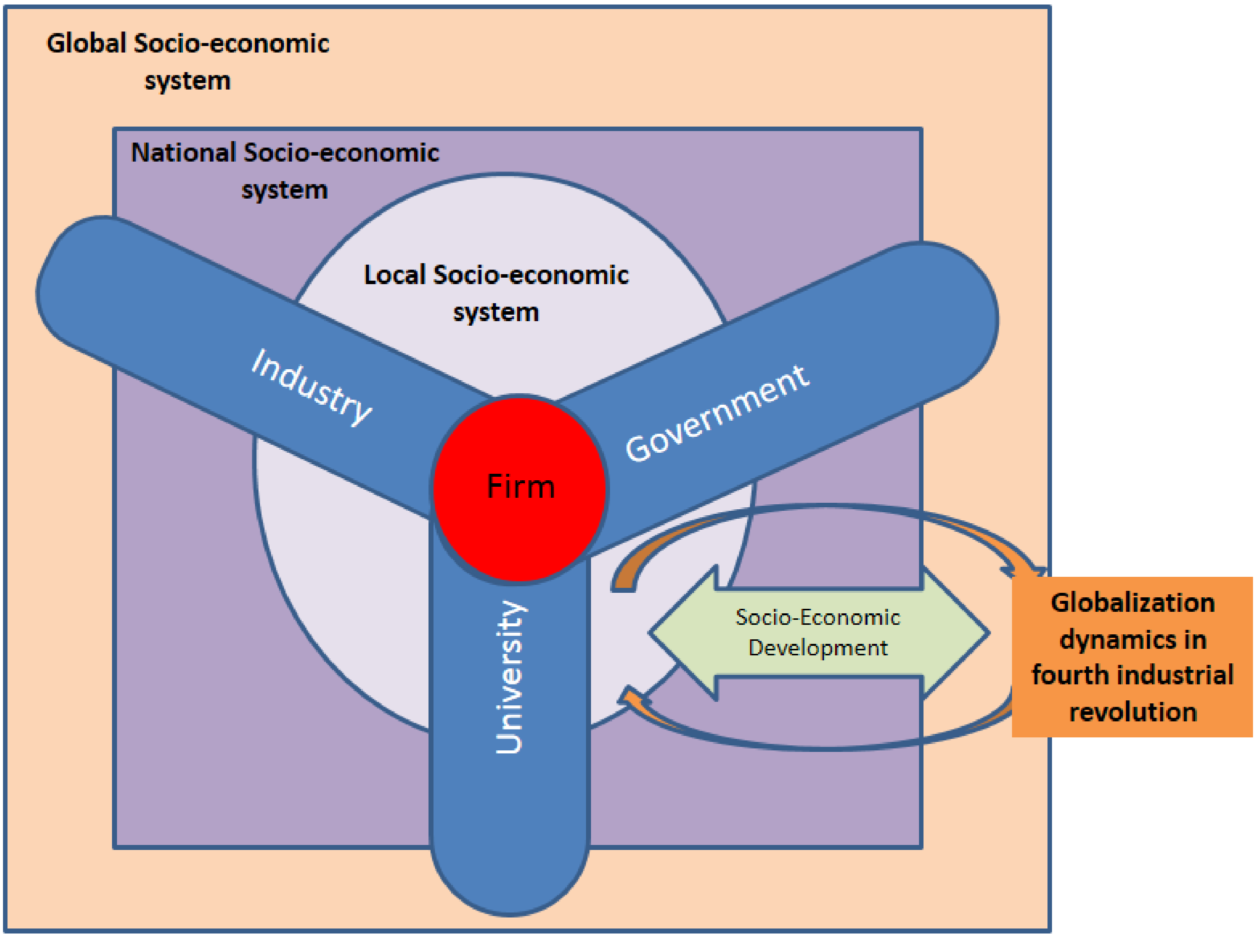
In this fourth industrial revolution context, a future direction of enrichment to the triple helix system of innovation is towards encompassing three profound analytical dimensions:

(a) The analytical dimension of the globalizing process, which unifies in spatial terms and repositions progressively all the innovation “helices”: the government policy, the industries, the academia, and the civil society;

(b) The analytical dimension of perceiving in consolidated terms the socioeconomic system’s dynamics, where all the economic, social, environmental, cultural, institutional, ideological and symbolic dimensions necessarily co-evolve; and

(c) The evolutionary micro-analytical dimension of the economic system, which attributes to the firm the core role in the developmental process in the fourth industrial revolution era (see Figure 12).

Figure 12: The triple helix in the era of the fourth industrial revolution



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